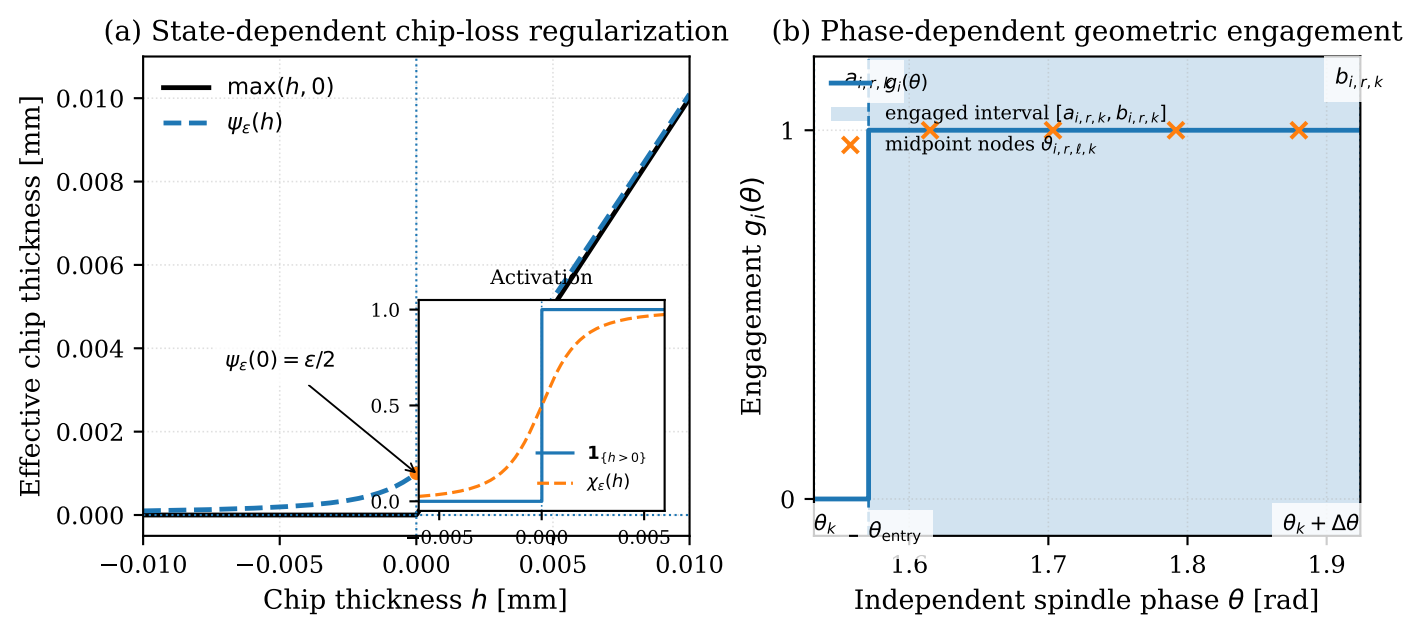
(a) State-dependent chip-loss regularization
max(h, 0)
ψε(h)
Effective chip thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
ψε(0) = ε/2
Activation
1.0
0.5
0.0
1{h > 0}
χε(h)
−0.005
0.000
0.005
−0.010
−0.005
0.000
0.005
0.010
Chip thickness h [mm]
(b) Phase-dependent geometric engagement
ai,r,k
gi(θ)
bi,r,k
engaged interval [ai,r,k, bi,r,k]
midpoint nodes ϑi,r,ℓ,k
Engagement gi(θ)
1
0
θk
θentry
θk + Δθ
1.6
1.7
1.8
1.9
Independent spindle phase θ [rad]